

Equation of motion

$$x[k+1] = x[k] + a_x[k]\{f_{dx}[k] + f_T[k]\} + \delta x_D[k],$$

Motion error

$$\delta x[k+1] = \delta x[k] + \Delta x_d[k] - a_x[k]\{f_{dx}[k] + f_T[k]\} - \delta x_D[k],$$

$\delta x[k] = x_d[k] - x[k]$, $x_d[k]$ is the desired motion trajectory, $\delta x_D[k]$ is the motion error caused by the disturbance, and $\Delta x_d[k] = x_d[k+1] - x_d[k]$ is the desired motion step size.

Measurement

(d -step delay)

$$\delta x_m[k] = \delta x[k-d] + n_x[k],$$

Control objective: $\delta x[k+1] = \lambda_c \cdot \delta x[k]; \quad 0 \leq \lambda_c < 1$

Ideal control effort (no measurement delay):

$$\bar{f}_{dx}[k] = a_x^{-1}[k]\{\Delta x_d[k] + (1 - \lambda_c) \delta \mathbf{x}[\mathbf{k}] - \delta x_D[k]\}$$

Motion error:

$$\delta x[k+1] = \delta x[k] + \Delta x_d[k] - a_x[k]\{\bar{f}_{dx}[k] + f_T[k]\} - \delta x_D[k] = \lambda_c \cdot \delta x[k] - a_x[k]f_T[k]$$

d -step measurement delay:

$$\delta \mathbf{x}[\mathbf{k}] = \delta x[k-d] + \sum_{i=1}^d \Delta x_d[k-i] - \sum_{i=1}^d a_x[k-i]\{f_{dx}[k-i] + f_T[k-i]\} - \sum_{i=1}^d \delta x_D[k-i]$$

Ideal control effort (with d -step measurement delay):

$$\begin{aligned} \bar{\bar{f}}_{dx}[k] = a_x^{-1}[k] \Big\{ & \underbrace{\left[\Delta x_d[k] + (1 - \lambda_c) \sum_{i=1}^d \Delta x_d[k-i] \right]}_{\Delta x_d^d[k]} + (1 - \lambda_c) \left[\delta x[k-d] - \sum_{i=1}^d a_x[k-i] f_{dx}[k-i] \right] \\ & - \underbrace{\left[\delta x_D[k] + (1 - \lambda_c) \sum_{i=1}^d \delta x_D[k-i] \right]}_{\delta x_D^d[k]} \Big\} \end{aligned}$$

Motion error:

$$\delta x[k+1] = \lambda_c \cdot \delta x[k] - \underbrace{\left\{ a_x[k]f_T[k] + (1 - \lambda_c) \sum_{i=1}^d a_x[k-i]f_T[k-i] \right\}}_{\delta x_T^d[k]}$$

Implementable control effort

$$\boxed{f_{dx}[k] = \hat{a}_x^{-1}[k] \left\{ \Delta x_d^d[k] + (1 - \lambda_c) \left[\delta x_m[k] - \sum_{i=1}^d \hat{a}_x[k-i] f_{dx}[k-i] \right] \right\}}$$

Motion error

$$\delta x[k+1] = \lambda_c \cdot \delta x[k] - a_x[k] [f_{dx}[k] - \bar{\bar{f}}_{dx}[k]] - \delta x_T^d[k]$$

$$a_x[k] \bar{\bar{f}}_{dx}[k] = \Delta x_d^d[k] + (1 - \lambda_c) \left[\delta x[k-d] - \sum_{i=1}^d a_x[k-i] f_{dx}[k-i] \right] - \delta x_D^d[k]$$

$$a_x[k] f_{dx}[k] = (1 + e_{a_x}[k] \hat{a}_x^{-1}[k]) \left\{ \Delta x_d^d[k] + (1 - \lambda_c) \left[(\delta x[k-d] + n_x[k]) - \sum_{i=1}^d \hat{a}_x[k-i] f_{dx}[k-i] \right] \right\}$$

$$\begin{aligned} a_x[k] f_{dx}[k] - a_x[k] \bar{\bar{f}}_{dx}[k] &= e_{a_x}[k] f_{dx}[k] + (1 - \lambda_c) n_x[k] \\ &\quad + (1 - \lambda_c) \sum_{i=1}^d (a_x[k-i] - \hat{a}_x[k-i]) f_{dx}[k-i] + \delta x_D^d[k] \end{aligned}$$

Parameter model

$$a_x[k] = a_{x_{det}}[k] + a_{x_{ram}}[k]; \quad a_{x_{ram}}[k] \approx a'_x[k] x_{ram}[k]$$

$$a_x[k+1] = \underbrace{\{a_{x_{det}}[k] + a_{x_{ram}}[k]\}}_{a_x[k]} + \delta a[k] + \underbrace{\{a_{x_{ram}}[k+1] - a_{x_{ram}}[k]\}}_{\delta a_{ram}[k]}$$

$$\begin{cases} a_x[k+1] = a_x[k] + \delta a_x[k] + \delta a_{ram}[k] \\ \delta a_x[k+1] = \delta a_x[k] \end{cases}$$

Parameter estimator:

$$\begin{cases} \hat{a}_x[k+1] = \hat{a}_x[k] + \delta \hat{a}_x[k] \\ \delta \hat{a}_x[k+1] = \delta \hat{a}_x[k] \end{cases}$$

Parameter estimation error:

$$\begin{cases} e_{a_x}[k] = a_x[k] - \hat{a}_x[k] \\ e_{\delta a_x}[k] = \delta a_x[k] - \delta \hat{a}_x[k] \end{cases}$$

Estimation error dynamics:

$$\begin{cases} e_{a_x}[k+1] = e_{a_x}[k] + e_{\delta a_x}[k] + \delta a_{ram}[k] \\ e_{\delta a_x}[k+1] = e_{\delta a_x}[k] \end{cases}$$

$$a_x[k-i] - \hat{a}_x[k-i] \approx e_{a_x}[k] - i \cdot e_{\delta a_x}[k] - \sum_{j=1}^i \delta a_{ram}[k-j]$$

$$\begin{aligned} a_x[k] f_{dx}[k] - a_x[k] \bar{f}_{dx}[k] &\approx e_{a_x}[k] \underbrace{\left\{ f_{dx}[k] + (1 - \lambda_c) \sum_{i=1}^d f_{dx}[k-i] \right\}}_{F_{dx}[k]} \\ &\quad - e_{\delta a_x}[k] \underbrace{\left\{ (1 - \lambda_c) \sum_{i=1}^d i f_{dx}[k-i] \right\}}_{dF_{dx}[k]} \\ &\quad - \underbrace{\left\{ (1 - \lambda_c) \sum_{i=1}^d (d+1-i) f_{dx}[k-i] \delta a_{ram}[k-i] \right\}}_{\delta x_{\delta a_f}^d[k]} \\ &\quad + \delta x_D^d[k] + (1 - \lambda_c) n_x[k] \end{aligned}$$

$$\begin{aligned} \delta x[k+1] &= \lambda_c \cdot \delta x[k] - F_{dx}[k] \cdot e_{a_x}[k] + dF_{dx}[k] \cdot e_{\delta a_x}[k] - \delta x_D^d[k] \\ &\quad - \underbrace{\left\{ \delta x_T^d[k] - \delta x_{\delta a_f}^d[k] + (1 - \lambda_c) n_x[k] \right\}}_{\varepsilon[k]} \end{aligned}$$

State equation

($\delta x_D^d \equiv 0$: disturbance not modelled)

$$\delta x_1[k+1] = \delta x_2[k]$$

$$\delta x_2[k+1] = \delta x_3[k]$$

$$\delta x_3[k+1] = \lambda_c \cdot \delta x_3[k] - F_{dx}[k] \cdot e_{a_x}[k] + dF_{dx}[k] \cdot e_{\delta a_x}[k] - \varepsilon[k]$$

$$a_x[k+1] = a_x[k] + \delta a_x[k] + \delta a_{ram}[k]$$

$$\delta a_x[k+1] = \delta a_x[k]$$

Output equation

$$\begin{cases} y_1[k] = \delta x_m[k] = \delta x_1[k] + \underbrace{n_x[k]}_{r_1[k]} \\ y_2[k] = \underbrace{a_x[k-d] + n_a[k]}_{a_{xm}[k]} \approx a_x[k] - d \cdot \delta a_x[k] + \underbrace{\left\{ n_a[k] - \sum_{i=1}^d \delta a_{ram}[k-i] \right\}}_{r_2[k]} \end{cases}$$

$$\mathbf{H} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & -d \end{bmatrix}, \quad \mathbf{y}[k] = \mathbf{H}\mathbf{x}[k] + \mathbf{r}[k]$$

Closed-loop Estimator

$$e_{\delta x_1}[k] = \delta x_1[k] - \delta \hat{x}_1[k]$$

$$e_{a_x}[k] = a_x[k] - \hat{a}_x[k]$$

$$e_{\delta a_x}[k] = \delta a_x[k] - \delta \hat{a}_x[k]$$

$$e_{y_1}[k] = y_1[k] - \hat{y}_1[k] = \delta x_m[k] - \delta \hat{x}_1[k] = e_{\delta x_1}[k] + \underbrace{n_x[k]}_{r_1[k]}$$

$$\begin{aligned} e_{y_2}[k] &= y_2[k] - \hat{y}_2[k] = a_{xm}[k] - \hat{a}_x[k] + d \cdot \delta \hat{a}_x[k] \\ &= e_{a_x}[k] - d \cdot e_{\delta a_x}[k] + \underbrace{\left\{ n_a[k] - \sum_{i=1}^d \delta a_{ram}[k-i] \right\}}_{r_2[k]} \end{aligned}$$

$$\delta \hat{x}_1[k+1] = \delta \hat{x}_2[k] + \ell_{11}[k] e_{y_1}[k] + \ell_{12}[k] e_{y_2}[k]$$

$$\delta \hat{x}_2[k+1] = \delta \hat{x}_3[k] + \ell_{21}[k] e_{y_1}[k] + \ell_{22}[k] e_{y_2}[k]$$

$$\delta \hat{x}_3[k+1] = \lambda_c \cdot \delta \hat{x}_3[k] + \ell_{31}[k] e_{y_1}[k] + \ell_{32}[k] e_{y_2}[k]$$

$$\hat{a}_x[k+1] = \hat{a}_x[k] + \delta \hat{a}_x[k] + \ell_{41}[k] e_{y_1}[k] + \ell_{42}[k] e_{y_2}[k]$$

$$\delta \hat{a}_x[k+1] = \delta \hat{a}_x[k] + \ell_{51}[k] e_{y_1}[k] + \ell_{52}[k] e_{y_2}[k]$$

Error dynamics

$$e_{\delta x_1}[k+1] = e_{\delta x_2}[k] - \ell_{11}[k] e_{\delta x_1}[k] - \ell_{12}[k] e_{a_x}[k] + \ell_{12}[k] d e_{\delta a_x}[k]$$

$$e_{\delta x_2}[k+1] = e_{\delta x_3}[k] - \ell_{21}[k] e_{\delta x_1}[k] - \ell_{22}[k] e_{a_x}[k] + \ell_{22}[k] d e_{\delta a_x}[k]$$

$$\begin{aligned} e_{\delta x_3}[k+1] &= \lambda_c \cdot e_{\delta x_3}[k] - F_{dx}[k] \cdot e_{a_x}[k] + d F_{dx}[k] \cdot e_{\delta a_x}[k] \\ &\quad - \ell_{31}[k] e_{\delta x_1}[k] - \ell_{32}[k] e_{a_x}[k] + \ell_{32}[k] d e_{\delta a_x}[k] - \underbrace{\varepsilon[k]}_{q_3[k]} \end{aligned}$$

$$e_{a_x}[k+1] = e_{a_x}[k] + e_{\delta a_x}[k] - \ell_{41}[k] e_{\delta x_1}[k] - \ell_{42}[k] e_{a_x}[k] + \ell_{42}[k] d e_{\delta a_x}[k] + \underbrace{\delta a_{ram}[k]}_{q_4[k]}$$

$$e_{\delta a_x}[k+1] = e_{\delta a_x}[k] - \ell_{51}[k] e_{\delta x_1}[k] - \ell_{52}[k] e_{a_x}[k] + \ell_{52}[k] d e_{\delta a_x}[k]$$

$$\mathbf{F}_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & \lambda_c & -F_{dx}[k] & dF_{dx}[k] \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$F_{dx}[k] = f_{dx}[k] + (1 - \lambda_c) \sum_{i=1}^d f_{dx}[k - i], \quad dF_{dx}[k] = (1 - \lambda_c) \sum_{i=1}^d i f_{dx}[k - i]$$

$$\mathbf{e}[k + 1] = \mathbf{F}_e[k] \mathbf{e}[k] - \mathbf{L}[k] \mathbf{H} \mathbf{e}[k] + \mathbf{q}[k] - \mathbf{L}[k] \mathbf{r}[k]$$